



MPAIRWE LAUBEN

DEVSECOPS · ML · AGENTIC AI
ENGINEER

ABOUT

DevSecOps engineer and applied ML practitioner. Co-founder of Iona Institute of Technology (ionatec.com). By day, CI/CD, cloud, and security posture at AIBOS Uganda. Outside: training models, building agentic systems on the Claude Agent SDK and LangGraph, and instrumenting MCP servers that expose tools to LLMs.

SKILLS

TRIVY

GRYPE

SNYK

COSIGN

SIGSTORE

SLSA

OPA / GATEKEEPER

KYVERNO

FALCO

VAULT

EXTERNAL SECRETS

POD SECURITY STANDARDS

NETWORKPOLICIES

GITHUB ACTIONS

ARGOCD

FLUX

TERRAFORM

PULUMI

MLFLOW

WEIGHTS & BIASES

DVC

FEAST

Kampala, Uganda +256 773 336 896

mpairwelauben75@gmail.com

WORK EXPERIENCE

AIBOS Uganda

Kampala

Feb 2024 – Present

DEVSECOPS ENGINEER

- Own CI/CD posture on GitHub Actions, cutting deploy time ~60% while adding SBOM generation, container scanning, and signature verification
- Run infrastructure-as-code (Terraform) across AWS, Azure, and GCP with policy-as-code gates on every plan
- Harden Kubernetes workloads with Pod Security Standards, NetworkPolicies, and supply-chain controls (cosign / Trivy)
- Wire Prometheus + Grafana + Loki dashboards with SLO-driven alerts and on-call runbooks
- Drive compliance posture (encryption, RBAC, secrets via Vault) across the cloud estate

GITHUB ACTIONS · TERRAFORM ·
KUBERNETES · TRIVY · COSIGN · VAULT ·
PROMETHEUS · GRAFANA · AWS

KUBEFLOW
RAY
TRITON
BENTOML
EVIDENTLY
SCIKIT-LEARN
TENSORFLOW
PYTORCH
PANDAS
NUMPY
CLAUDE AGENT SDK
LANGGRAPH
MODEL CONTEXT PROTOCOL
VERCEL AI GATEWAY
AI SDK V6
QDRANT
PGVECTOR
BRAINTRUST
PROMPTFOO
LANGSMITH
DEEPEVAL
GUARDRAILS
AWS
GCP
AZURE
VERCEL
KUBERNETES
DOCKER
HELM
PROMETHEUS
GRAFANA
LOKI
OPENTELEMETRY
ISTIO
POSTGRESQL
PGVECTOR
MYSQL
MONGODB
REDIS
QDRANT
SNOWFLAKE
FIREBASE
SUPABASE
NEXT.JS
REACT
TYPESCRIPT
TAILWIND
SHADCN/UI

Ionatec · Iona Institute of Technology

Kampala

Feb 2023 – Present

CO-FOUNDER · ML & AGENTIC ENGINEER

- ▶ Co-founded Ionatec (ionatec.com) — an engineering institute and applied-AI studio training East African engineers on real production stacks
- ▶ Ship LLM agents on the Claude Agent SDK + LangGraph: tool-use, retrieval, multi-agent orchestration, and evals
- ▶ Build RAG pipelines on Qdrant / pgvector with hybrid retrieval, rerank, and Braintrust/Promptfoo eval harnesses
- ▶ Deliver classical ML for classification, regression, and ranking — with monitoring, drift detection, and retraining loops
- ▶ Run the engineering curriculum and mentor residents from prototype through production

CLAUDE AGENT SDK · LANGGRAPH · MCP · QDRANT · BRAINTRUST · PYTHON · FASTAPI · NEXT.JS · SCIKIT-LEARN

Makerere University

Kampala

Jan 2022 – Present

STUDENT SOFTWARE ENGINEER

- ▶ Built TaskApp — a Kotlin task management app with real-time Firebase sync
- ▶ Developed scalable cross-platform apps using React Native and Flutter
- ▶ Implemented secure JWT and biometric authentication flows
- ▶ Delivered a fintech mobile app with digital payments, rewards, and transaction tracking
- ▶ Integrated third-party payment APIs and designed scalable backend architecture

KOTLIN · REACT NATIVE · FLUTTER · FIREBASE · JWT · FINTECH

MOTION

NODE.JS

FASTAPI

SPRING BOOT

TRPC

FLUTTER

REACT NATIVE

KOTLIN

SWIFT

FIREBASE

APP STORE / PLAY

LINKS

 LINKEDIN
mpairwe-lauben-3a1461344

 GITHUB
github.com/mpairweLandwind

 IONATEC
ionatec.com

LANGUAGES

English (Fluent)

German (Basic)

PERSONAL DETAILS

DATE OF BIRTH
22 October 1995

NATIONALITY
Ugandan

LOCATION
Kampala, Uganda

REFERENCES

Dr. Kimbugwe Nasser
Lecturer & Academic Supervisor
Makerere University, School of Computing & IT

Kalema Martin
Supervisor
AIBOS Uganda

Dr. Mark Magumba
Senior Lecturer
Makerere University, School of Computing & IT

Fahadi Nguyenguye
Technical Lead
PAHAPPA LIMITED

PAHAPPA LIMITED

Kampala

Jun 2022 – Jan 2024

WEB INFO-NET LIMITED

Kampala

Jun 2018 – Aug 2022

ZALONIDENTALHUB

Kampala

Apr 2022 – Present

EDUCATION

DEVOPS INTERN → ENGINEER

- ▶ Built enterprise applications with Spring Boot (MVC, Data JPA, Security, Cloud)
- ▶ Used Hibernate ORM/JPA with MySQL, PostgreSQL, Oracle, and MongoDB databases
- ▶ Designed Microservice and Monolithic architectures based on project requirements
- ▶ Developed RESTful APIs and event-driven systems for high-throughput workflows

SPRING BOOT · JAVA · HIBERNATE ·
MYSQL · POSTGRESQL · MICROSERVICES ·
REST API

APPLICATION DEVELOPER INTERN & TRAINER

- ▶ Developed mobile apps using Swift, Java, and Kotlin for iOS and Android platforms
- ▶ Built an Odoo-based Employment Management System for HR automation
- ▶ Implemented secure authentication and third-party API integrations
- ▶ Provided Level 1 technical support, troubleshooting, and end-user training

SWIFT · JAVA · KOTLIN · OD00 · IOS ·
ANDROID

LEADING DEVELOPER & MAINTAINER

- ▶ Led full SDLC for dental practice management software from conception to delivery
- ▶ Built C#/VB.NET desktop systems for inventory and patient management
- ▶ Developed a Flutter mobile app with e-commerce, appointments, and booking features
- ▶ Ensured seamless desktop-mobile data interoperability
- ▶ Led a team of developers, defining software architectures and coding standards

C# · VB.NET · FLUTTER · DESKTOP ·
MOBILE · TEAM LEAD

Makerere University

2024 – Expected
2027

CGPA 4.31

Bachelor of Science in Software Engineering

- ▶ Specialising in AI/ML and Enterprise Application Development
- ▶ Integrated Supabase, MongoDB, Cloudflare, and Firebase for scalable storage
- ▶ Implemented RBAC and encryption for secure multi-role systems
- ▶ Built a full HRMS system with advanced role and permissions management
- ▶ Worked on IoT and Embedded Systems research projects
- ▶ Optimised PostgreSQL queries for high-performance data retrieval

Makerere University

2019 – 2023

Bachelor of Science in Computer Engineering

- ▶ Transitioned to the Software Engineering programme
- ▶ Studied Digital Electronics and hardware design principles
- ▶ Hardware repair, maintenance, and systems diagnostics

Uganda Technical College – Lira

2016 – 2018

CGPA 4.77

National Diploma in Information Communication Technology

- ▶ Database Management and Web Development
- ▶ Internet technologies and networking fundamentals
- ▶ RBAC and Encryption implementation
- ▶ Java programming and object-oriented design
- ▶ Networking and Server Configuration

Boston High School – Mpala

2011 – 2015

Uganda Certificate of Education (UCE)

- ▶ Graduated in Top 5% of class
- ▶ Varsity Debate Team Captain

CERTIFICATIONS



PLANNED	AWS Certified DevOps Engineer · Professional	— Amazon Web Services
PLANNED	Certified Kubernetes Security Specialist	— CNCF
PLANNED	Certified Kubernetes Administrator	— CNCF
PLANNED	HashiCorp Certified · Terraform Associate	— HashiCorp
PLANNED	AWS Solutions Architect · Associate	— Amazon Web Services
PLANNED	Google Cloud Professional ML Engineer	— Google Cloud